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The application of artificial intelligence in the sonography profession: Professional and educational considerations

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Abstract

The integration of artificial intelligence (AI) technology within the health industry is increasing. This educational piece discusses the implementation of AI and its impact on sonography. The authors investigate how AI may influence the profession and provide examples of how ultrasound imaging may be enhanced and innovated by integrating AI technology. This article highlights challenges related to the application of AI and provides insight into how they could be addressed. The critical distinction between the role of a sonographer and the reporting specialist in the context of AI is highlighted as a key issue for those developing, researching, and evaluating AI systems. A key recommendation is for the sonography community to address ultrasound education, particularly how AI knowledge could be incorporated into university education. This is an important consideration that should be extended to practising professionals as they may be involved in evaluating the efficiency and methodologies used in new research that may incorporate AI technologies.

Keywords

AI technology, ultrasound imaging, sonographer, practice, education

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Introduction

Artificial intelligence (AI) and AI-based technologies are increasingly prevalent in daily life. From search and media recommendations (Google, Netflix and Spotify), face and speech recognition, to the ongoing emergence and development of autonomous vehicles. A globally ageing population and increasing financial burdens of chronic disease have challenged health administrations and enterprises to transform healthcare delivery, resulting in the largest investment in AI since 2016.[1] This includes drug development technologies, patient triage functions, diagnosis, patient informatics technologies and various applications in the field of medical imaging.[2] Like facial recognition technology, advances in AI algorithms have led to the development of a range of image classification and pattern recognition systems that perform many of the functions of medical imaging professionals.[3]

The literature investigating AI in the medical imaging field has exploded in the last few years. From 2016 to 2021, abstracts listed on PubMed referencing keywords 'artificial intelligence' and 'medical imaging' have more than doubled when compared to the corresponding previous 5-year period.[4] Already, many of these technologies are approved for human use. The American College of Radiology Data Science Institute®[5] tracks the number of Federal Drug Administration (FDA) approved AI systems and lists more than 100. Most of these involve plain radiography, computed tomography (CT), and magnetic resonance imaging (MRI); however, several involve ultrasound (US) images.

US is a valuable and ubiquitous diagnostic tool utilised by many healthcare professionals, from highly skilled

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technologists such as sonographers to those using the technology for a singular diagnostic or therapeutic task. The emergence of AI in US is likely to play a different role for different groups and settings to meet disparate needs. AI applicability for point-of-care US (POCUS) may focus on acquiring standard planes with a greater emphasis on technical skills. In more specialised settings, AI may assist in further developing emergent technologies such as improving border detection during echocardiography or reducing intraoperator variability of two-dimensional (2D) shear wave elastography.[6]

The integration of AI technologies into the sonography workflow will not be without challenges. Sonographers, unlike other medical imaging professionals, are directly coupled with the technology. The success of the examination is highly dependent on the operator's skill and experience; sonographers adapt in real-time to changing clinical and patient cues, simultaneously evaluating a rapidly changing image to optimise patient care. Sonographers are uniquely positioned to humanise technology, providing patient-centred care, a key tenant of modern medicine.[7] A recent report from the European Parliament Committee on Legal Affairs[8] stated that 'human contact is one of the fundamental aspects of human care'. How AI can develop to perform these types of seemingly intuitive yet complex tasks efficiently remains unclear. There are also other important questions concerning privacy and ethics that need consideration.

Unlike other articles on the topic, this educational piece will focus on implementing AI and its impact on the sonography community now and into the future. The authors provide a basic overview of the key terminology encountered in the literature surrounding AI, including several examples of how the diagnostic power of US imaging may be enhanced as elements of AI are embedded. However, the discussion will extend beyond just the US image and investigate how the technology may influence the profession at large and how it has innovation potential. Finally, it addresses the critical issue of US education and how research into its applicability should be a key consideration for practising professionals.

What is artificial intelligence

The concept of AI and the potential for machines to be automated can be traced to Greek mythology. However, the widely celebrated English mathematician, computer scientist and code-breaker Alan Turing is considered the modern father of AI.[9] In 1950, Turing[10] published the seminal paper 'Computing Machinery and Intelligence', posing the question, 'Can machines think?'. The International Standardisation Organization[11] provides a current definition of AI as the *capability of an engineered system to acquire, process and apply knowledge and skills*. In addition to the definition of AI, several other terms prevail in the literature. Here, we explain some of the central terms and common concepts relevant to AI.

Machine learning

Machine learning (ML) is an algorithm or system that can learn from data without relying on specific rules-based programming. The machine 'learns' from the input data, not from what the human thinks or believes the data shows.[12] For health scientists, analysing data aims to extract new information about the system under investigation or make predictions when particular measurements or observations are identified. The traditional approach to solving these problems is to use statistical modelling techniques to look for patterns in the data.[13] Statistical modelling deals with finding a relationship between variables to predict an outcome using a mathematical rules-based approach. For example, simple linear regression, a form of statistical modelling, attempts to estimate the function of a straight line between two variables.[14] This process is mathematically intensive and requires the researcher to understand the relationships between variables before input and analysis. ML is free of these limitations, focusing primarily on the best method to predict an outcome from all available inputs within intrinsic ML model assumptions.[15] ML works through a series of iterations in which the computer looks for patterns in the data. Since datasets are often large, the predictive power of these models is very strong.[16]

The end goal of both traditional and ML analytical approaches is the same; however, the formulation of the two are significantly different.[15] Traditional statistical models can utilise much smaller sample sizes when correct assumptions are met. In contrast, ML algorithms are data-hungry and, in some scenarios, may require millions of observations to reach acceptable performance.[16] The power of the ML approach in medical imaging is its ability to offer many new and exciting correlations that may not have been obvious before data analysis. Some may even result in unexpected and concerning findings; for example, readers may be aware of recent reports of the ability of AI to identify an individual's race from chest radiographs.[17]

Deep learning and convolutional neural networks

A conventional ML approach requires significant engineering and domain expertise to programme a system to classify patterns in the input data. Modern methods now utilise techniques in which features are no longer 'hand-crafted' into the system (Figure 1). Instead, the computer learns specific features in the data to correctly classify it. An artificial neural network (ANN) is an analysis algorithm composed of connected nodes, which model organic neurons in

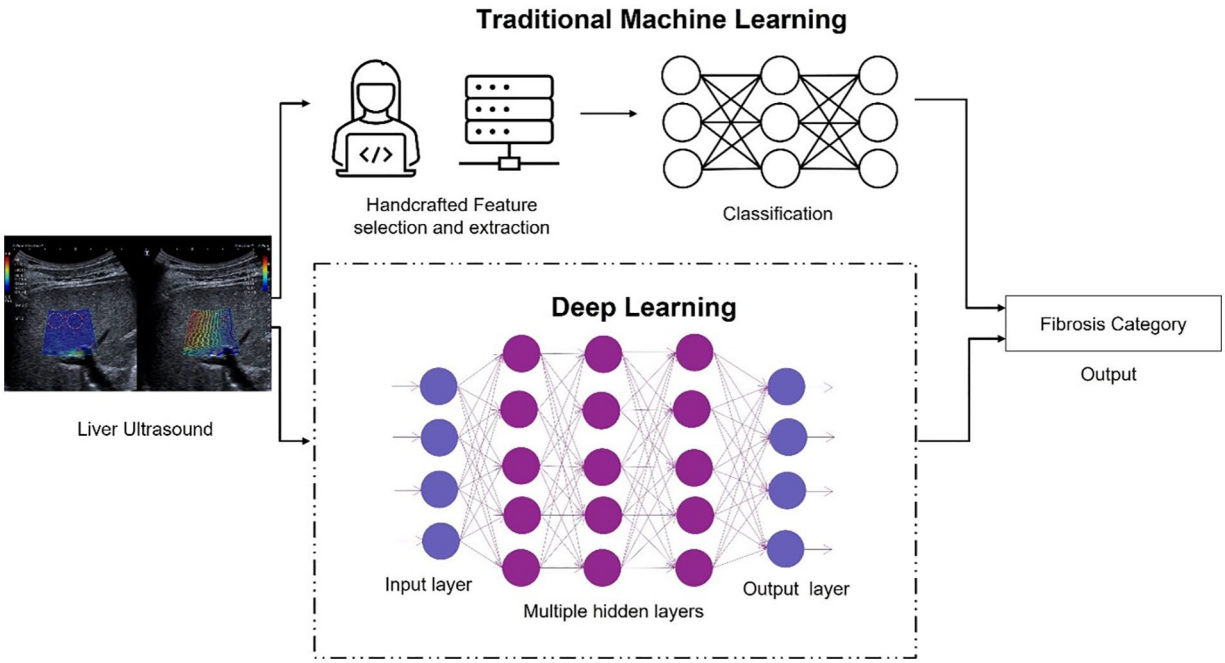

Figure 1. Hypothetical schematic illustration of Traditional Machine Learning versus Deep Learning. In this example, the authors demonstrate how data acquired from shear-wave elastography can be used to predict the liver fibrosis category.

a brain. Like synapses in a biological brain, each node can transmit data to other nodes. Deep learning (DL) involves using artificial cognition afforded by layered ANN nodes, in the order of hundreds to millions, to self-learn complex relationships and representations within input data and improve predictive analytical ability.[18,19]

One revolutionary development of DL relevant in medical imaging applications is Convolutional Neural Networks (CNN). These multi-layered ANNs explicitly assume inputs are images, making them particularly adept at the classification tasks associated with spatial information in medical images.[20] Initial layers in the CNN contain filters to detect basic image features such as edges, abstract shapes or shading. In contrast, subsequent layers will focus more on the clinical diagnostic task of the algorithm, such as anomaly and pathology detection. The clever part of these systems is their ability to be trained via supervised learning. During the training phase, relevant features are stored, and errors are fed back within the network to optimise the predictive power of the network as more data are added.[21] The layers within these types of neural networks are called 'hidden layers' because the system will decide on its own which features are the most important to gain the highest probability of success. However, while the mathematical architecture governing these networks may be easily explained, it is difficult to define which patterns in the data the network used to draw its conclusion.[22] As a direct result, there are some criticisms and concerns around the degree of trust we place in these DL systems; even researchers in the field do not fully understand them. In addition, these types of networks require significant computational time particularly during the analysis of large and diverse data sets.

Artificial intelligence and medical image analysis

The literature is currently overflowing with articles on the impact of AI on diagnostic imaging, from widespread industry disruption to more considered commentaries around the industry's evolution to a more patient-focused one.[23] DL algorithms are not susceptible to human cognitive bias or environmental conditions such as distraction and fatigue and, therefore, can be at a significant advantage.[24] However, a recent review of 100 commercially available AI products in radiology found that most (64) lacked peer review evidence. Only 18 demonstrated potential clinical impact, indicating that the development of AI in medical imaging is still in its infancy.[25] While some algorithms tested in controlled environments may perform well, AI can struggle to reliably distinguish medical images when presented with inappropriate inputs, such as images of flowers or cars, highlighting the need for careful oversight of systems.[26]

DL system development requires data that can be divided into training, validation, and test sets. This process provides a basis for learning, proof of accuracy and a means to assess if results can be generalised more widely. Training datasets include images that are appropriately labelled or contain segmented regions of interest or anatomy. Unfortunately, the availability of labelled or segmented datasets for modalities such as MRI and CT far exceeds that of US. US

imaging also poses unique problems as it lacks standardised planes and is highly influenced by noise, artefacts and field of view limitations.

Impacts and applications of artificial intelligence in sonography

When considering the full impact of AI on sonography, we break our narrative into some of the key aspects relevant to the sonographer's role. These practice areas include diagnosis and detection, image acquisition, quality control and image optimisation, education/training and research.

Diagnosis and detection

Detecting and diagnosing pathology is a primary role of medical imaging and is the aim of many AI-enabled systems. Several authors have detailed some of the emerging applications specific to US.[27,28] Within this section, AI-based systems focus on classification and segmentation. Classification systems, for example, can provide the probability of the malignant potential of a lesion identified on screening ultrasound, such as classifying liver, ovarian, breast or thyroid masses. One commercial example is Samsung's S-Detect™, which provides a classification of suspicious breast lesions using the American College of Radiology Breast Imaging Reporting and Data System (BI-RADS®) Atlas.[29] Segmentation offers a means for accurate border delineation, detecting specific areas of interest within the scan plane. One novel application developed by Looney et al.[30] and trained on a large three-dimensional (3D) US dataset, OxNNet, can segment the placenta in the first trimester. The placenta volumes in this work are correlated with birth weight, and the system demonstrates a high predictive power to identify small for gestational age babies. Zhang et al.,[31] using over 14,000 echocardiogram images, developed a system that could segment the various cardiac views, provide quantification data on cardiac structure and function and disease detection such as hypertrophic cardiomyopathy, cardiac amyloid and pulmonary hypertension.

Examples like these and others will undoubtedly challenge the current paradigm of image interpretation and reporting into the future. Their ability to speed up reporting, reduce intraobserver variability and provide novel diagnostic yields have exciting benefits to patient care and outcomes. Ultrasound examinations that already utilise structured reporting nomenclature will see the most use. Systems that quickly and accurately classify an identified lesion into established BI-RADS and Thyroid Imaging Reporting & Data System (TI-RADS™) frameworks may reduce unnecessary duplication between the sonographer and radiologist and improve workflow.

It is important to note that these systems do not acquire the image and are blind to some of the critical clinical cues available to the sonographer at the time of imaging. Such variables can include a patient's clinical history (family or surgical) or be elicited during the ultrasound examination (pain, physiological responses and positional related changes). These factors are things sonographers integrate to optimise imaging but are currently not routinely collected or recorded and can be quite varied, making them difficult for inclusion in an AI training algorithm. Further complicating the application of AI diagnosis can be the lack of an accepted ground truth or specified finding concerning some conditions, which may not be a binary true/false statement, but potentially on a continuum or likelihood or severity.[32] For example, establishing a positive Murphy's sign to strengthen the clinical picture of acute cholecystitis requires ensuring the elicited pain is due to the tenderness of an inflamed gallbladder and not anatomically similar yet clinically divergent causes, and AI may struggle in this differentiation.

Quality control and image optimisation

The process of acquiring standard planes, increasing measurement accuracy and reducing intraoperator variability is another potential application of AI. There are already several commercially available systems that assist the acquisition of normal foetal biometric planes, including Mindray's Smartplanes®, Samsung's 5D CNS + ®, GE Healthcare' SonoCNS® Foetal Brain (Figure 2(a)) and Canon Medical's Biometry Assist (Figure 2(b)). ScanNAV assist (Intelligent Ultrasound) available on some GE systems recognises the 20 views recommended by the International Society of Ultrasound in Obstetrics and Gynaecology for second trimester foetal screening.[33] Other designs include the Philips HeartModel system, an automated ML algorithm that can quickly measure dynamic LV and LA volumes and accurately analyse ejection and filling parameters (Figure 3).[34]

The apparent potential is for the novice practitioner, particularly in educational settings. However, more experienced sonographers should also consider technologies that improve intraoperator consistency, even without a noticeable workflow improvement benefit. In the case of obstetric ultrasound, accurate serial measurements throughout pregnancy are critical to ensure appropriate care and management. Variability in biometric measurements can raise the possibility that foetal development may need further investigation. In contrast, deviation from expected growth vectors on longitudinal scans due to interoperator variability can result in unnecessary examinations and anxiety. In cardiac ultrasound, AI-enabled systems that utilise the entire 3D transthoracic data set rely less on 2D image quality and demonstrate excellent agreement with cardiac MRI imaging.[35] These systems will likely see benefits from referring clinicians evaluating cardiac physiology across multiple modalities.

The lack of ionising radiation, increased portability and affordability makes US an imaging modality with great potential for screening programmes for some conditions.

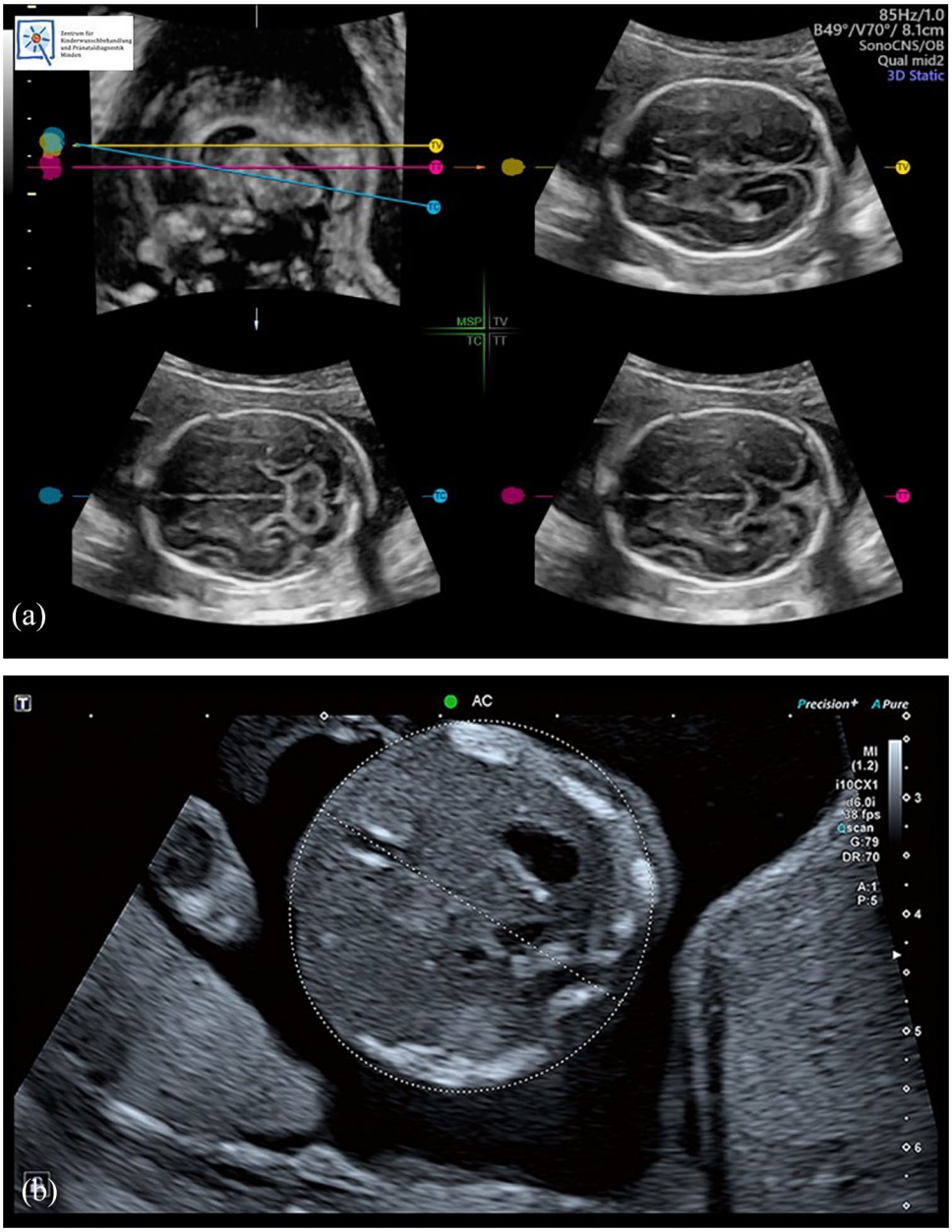

Figure 2. Examples of foetal biometry plane detection. (a) SonoCNS® Foetal Brain (used with permission of GE Healthcare (Chicago, IL, USA); (b) Biometry Assist (used with permission of Canon Medical ANZ (Castle Hill, NSW, AU).

However, the availability of an expert user to ensure a lack of false results can make the logistics of implementing such programmes a barrier. Some countries have adopted the use of US in universal screening to diagnose developmental dysplasia of the hip (DDH) in infants, as the long-term public health implications of an undiagnosed case can outweigh the cost of the screening programme. These examinations are performed either by orthopaedic surgeons or by sonographers using strict technical parameters to ensure consistency. However, they can still lead to over-diagnosis and over-treatment,[36–38] and there is some debate about when and how best to perform screening or surveillance for the condition.[39] AI has been demonstrated to accurately analyse US 3D datasets of infant hips and predict the likelihood and severity of DDH,[40] removing much of the operator variability from the examination.[41] The Medo Hip (Medo AI, Alberta, Canada) application is now an FDA-approved cloud-based AI product that has the potential to transform the role of US in diagnosing DDH.

Image quality is another consideration and is a function of the sonographer's ability and experience, equipment quality and various patient factors. As Kim[42] describes, even competent practitioners may fail to produce diagnostic images simply due to patient compliance or habitus. Inter- and intra- vendor variations arise from subtleties in image processing, types of transducers, technical set-ups of

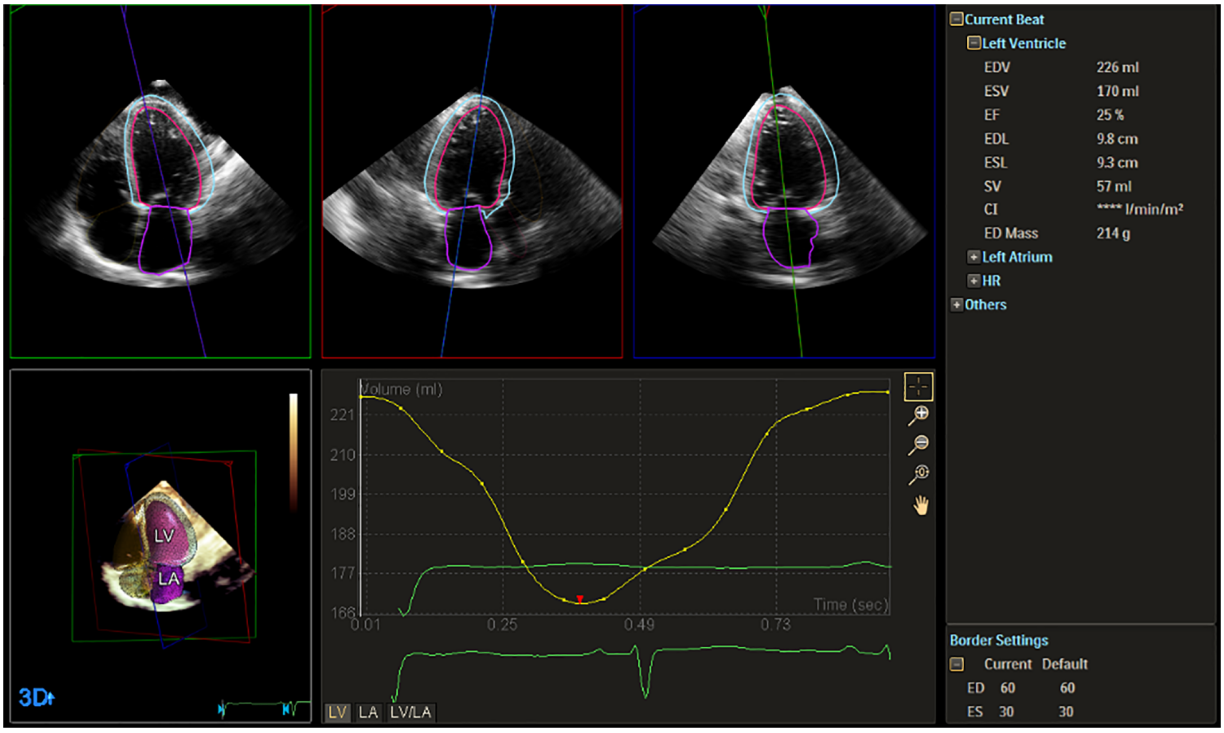

Figure 3. Dynamic HeartModel[A.I.] (used with permission of Philips Australia and New Zealand (Macquarie Park, NSW, AU) display showing the automatic alignment of dynamic contours across multiple views together with the volume waveform.

machines such as image compression and contrast across platforms. For example, the same machine model may have different dynamic ranges set for vascular presets compared to general imaging presets, influencing image quality and how AI interprets it for training or clinical use. Differences between high-end cart-based systems versus small form POCUS style machines are another potential image quality variation area.[43] Poor performing ultrasound equipment such as those with crystal dropout can also impact the recorded image quality. To date, many published AI studies have utilised a limited range of US machines and institutions, potentially introducing bias. Therefore, careful validation of findings is required before more widespread use.

Image acquisition

It is crucial to recognise the unique nature of sonographic practice in comparison to other healthcare professions and discuss if AI tools can replicate psychomotor skills and the human interactions that work in tandem. Unlike other medical imaging modalities where the technologist remotely controls image acquisition, a sonographer examines and captures images via direct patient contact. Sonographers develop skills creativity both from an image interpretation perspective and via improved physical dexterity.[44] Sonographers rely on varying patient positions and breathing techniques while simultaneously adapting their bodies and transducer positions to optimise image quality. Interestingly, AI technologies are also being developed by directing probe manipulation elements of the task. Caption Health, a company integrating AI solutions for cardiac US, will challenge the current roles of sonographers.[45] Their system uses 'artificial intelligence to provide real-time guidance and diagnostic quality assessment of images', claiming significant benefits for novice practitioners, 'even those without prior US experience'.[46] This system, currently only specific to echocardiography, provides on-screen guidance to assist the operator in obtaining optimal on-axis cardiac views.

Removing the sonographer completely and developing a purely autonomous system would require a significant advancement in robotic technology, image interpretation and human speech synthesis. However, investigations into automatic robotic US are not beyond reality, particularly in response to the need to enable physical distancing during the COVID-19 pandemic.[47] Advancements in robotic surgery are possibly analogous, and there has been tremendous advancement in this field. However, even the proponents of robotic surgery report that its success 'remains dependent on the surgeon's skill'.[48] One obvious distinction is that the robotic surgery patient is usually anaesthetised, and there is no requirement for concurrent communication. One area where technology in this space is likely to benefit is ultrasound-guided treatment planning during radiation therapy (RT). Some commercial RT systems (Clarity®–Elekta, Crawley, United Kingdom) already contain a transperineal

ultrasound imaging guidance system to assist dose delivery to the prostate. These systems still require an operator with sonography experience to position the probe and optimise image quality. AI-based systems designed to detect the prostate on transperineal images and assess the quality of images in real-time are currently in development.[49]

Beam optimisation

Fundamental improvements in image quality are also likely to accelerate as beamformer design implements AI technologies. Modern ultrasound systems are now moving away from conventional beamforming methods, which rely on the careful control of geometric and acoustic characteristics of the transmit beam. This method results in significant compression and truncation of received data. Improvements in computer hardware and data transfer rates now allow systems to receive and analyse the entire bandwidth of received signals, allowing virtual beamforming. Software algorithms can now virtually focus the beam, dramatically improving spatial resolution throughout the field of view, without significant data or temporal loss. This 'new paradigm'[50] of US diagnostics will likely advance further as innovative DL algorithms are applied directly to the received data channel.[51] This is already occurring in other modalities such as CT and MRI with the implementation of AI to reduce image noise resulting from radiation scatter or motion artefacts.[52] Such technology could be beneficial in the field of shear-wave elastography, which has limitations in terms of the quality of the underlying image and variability due to user region of interest selection.[53] Via the analysis of the entire received raw data, AI technologies could alert the user to sample acquisition outside set parameters, such as greater than or less than two standard deviations, indicating that the measurement should be repeated.[6]

Sonography education

The emergence of AI-based technology in the sonography profession will likely increase the education demand to include AI perspectives. For university education, the first consideration is evaluating how the new technologies may disrupt teaching and learning, specifically the scope of the content delivered and assessment. Training environments could be used as platforms to inform novice practitioners about possible disruptions to future practice, particularly around the potential benefits and challenges they are likely to bring to the profession. At the same time, it is crucial sonographers understand how the broader healthcare industry is responding to AI-based approaches and how it could affect future roles. Specific areas that could be embedded in sonography university education include the following.

AI and ML fundamentals. Contemporary healthcare literature highlights key areas that could form part of curriculum development. These include fundamental concepts of AI and ML, quality and safety, understanding the current applications of AI and how it will impact clinical workflows.[54] This will allow sonographers to critically analyse emerging technologies and engage with other professionals in furthering discussions of AI.

AI impact in sonography. Specific to sonography curriculum enhancement, there needs to be careful consideration of specific areas where AI technologies will impact sonographic practice, such as those highlighted above. Ultrasound is considered highly user-dependent; building a relevant curriculum that embeds an understanding of AI systems that can address these concerns should be a priority. Educators may strengthen the practical components of teaching and learning by embedding AI-based tools students may encounter in real-world US scanning.

AI privacy, ethics and legal aspects. The need for greater ethical and legal accountability in applying *AI in healthcare cannot be over-emphasised.* Pesapane et al.[55] argues that *even if AI can mimic human brain neural networks, (they) think differently than humans.* One of the priorities in sonography education must be engaging students in conversations regarding the ethical and moral application of AI in healthcare. Some of the concerns highlighted in the current literature include safety and transparency in the use of data, data privacy and intellectual property fairness and biases that AI algorithms may introduce. While sonography education already addresses legal and ethical principles related to patient care, further cultivating students' ability to discern potential algorithmic limitations by embedding examples of unethical use of data for research or training of AI models is of utmost importance.

Research and development opportunities

Tackling some of the challenges in the implementation of AI will require research and ongoing development and evaluation of AI-based technology. This includes projects with a clinical and educational focus. For example, a potential research area in sonography education could involve implementing AI technologies to improve skill acquisition and student engagement and augmenting current teaching and learning technologies such as virtual reality. As discussed earlier, computer-assisted/directed positioning of the probe in US technology could be such a way to improve simulation in US education.

As experts in US imaging, qualified sonographers are well placed to operate at the front line in AI research activities. Medical imaging companies and research laboratories have already employed sonographers to assist in the supervised learning phase of DL algorithm development. The recently published AI guidance to imaging professionals by the UK Society of Radiographers[56] identifies several

key priority areas. These include aspects of the impact of AI on patient care outcomes, clinical decision-making and potential new career opportunities for medical imaging professionals.

The clinical evaluation of these systems in non-standard clinical environments will be another exciting field of research. In challenging clinical environments, sonographers employ novel heuristics such as the cardiac sonographer adapting to patient positioning challenges in the intensive care unit (ICU), the paediatric sonographer imaging a child while clutching their mother or the obstetric sonographer managing a complicated foetal lie. There is a real possibility training datasets may exclude these types of examinations, causing problems when the system is faced with these everyday situations. Understanding these types of challenges should be a focus of research in the field.

Summary and recommendations

Emerging literature highlights some of the key areas in which AI technologies are poised to impact the sonography profession. However, there are still differing opinions on the future abilities of AI and how they might affect clinical practice, education and research. One of the challenges for AI in the field will be adapting the technology to the novel and unexpected and advancing it to utilise common sense reasoning. In the examples above, we have separated some of the key tenets of how AI may influence the sonography profession. One additional consideration is the potential danger of reducing the modality into a series of easily definable parts. Experienced sonographers creatively combine multiple elements into what some describe as an art form.[57] The possibility of AI integration resulting in a decoupling of these factors, generating over-reliance in some scenarios or a reluctance to use if systems only demonstrate a benefit on the optimal patient, should be considered when applying these to the real world.

If sonographers are to deal effectively with changes in clinical processes as a result of AI-based technologies;

- There is a need to embed AI and ML concepts into the curriculum, with ongoing content refinement informed by evidence of clinical applications and emerging research specific to sonography and practice.
- While fundamentals of AI and ML may be systematically introduced in university education, clinical sonographers also need to understand the applications, potential benefits and associated risks of their clinical applications.
- To ensure the critical analysis of patient safety and legal and ethical issues that may arise, practitioners should engage with the broader healthcare community; this will ensure any implementation of technologies is focused first on patient care and outcomes.

Conclusion

The practical implementation of AI methods and assessing the correlation between automated results and clinical outcomes is vital for future research. It is paramount to address how the current workforce can be educated on the impact and potential of recent AI developments and their limitations. While there are tremendous opportunities, there are ethical and practical considerations that will undoubtedly impact sonographer roles. It should not simply be enough to trust the 'black box' algorithms built into the software; sonographers must have some understanding of how they are designed and evaluated before applying them in real clinical environments. Finally, certain unique aspects of the sonographer's role are unlikely to be augmented by current AI-enabled technologies. Sonographers are the critical conduit between the patient and their machines, employing an array of adaptive patient interactions to optimise imaging and patient care outcomes.

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